

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (EC/ME/CL/CE/IT/EE)

December 2011

CL102 Environmental Sciences

Date: 14.12.2011, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Define the term Environment. Briefly describe physical components of it. [03]
(b) Give the composition of atmosphere in briefly. [04]
- Q - 2 (a) Explain Component parts of Environment and explain lithosphere in detail. [04]
(b) Define Noise Pollution. Describe Sources of Noise pollution. [05]
(c) Enlist parameters of water quality standards for drinking water in India specified by India. [05]

OR

- Q - 2 (a) Describe Structure of Atmosphere in detail. [04]
(b) Explain the term "Environmental Degradation" and discuss the role of advanced Technology in the degradation of environment. [05]
(c) Describe effects of Air Pollution on plant. [05]
- Q - 3 (a) Give the classification of air pollutants based on source. [04]
(b) Enlist and explain any two sources of water pollution. [05]
(c) What is 'Acid Rain'? Write a brief note on effects of acid rain on environment. [05]

OR

- Q - 3 (a) Enlist the common air pollutants stating their permissible limits as per ambient air Quality standards. [04]
(b) Enlist all current global environmental problems and Write a note on Global Warming. [05]
(c) Give the classification of water quality parameters and explain any two. [05]

SECTION - II

- Q - 4 (a) Define 'Ecology'. Write environmental significance of Hydrological Cycle. [03]
(b) Enlist the components of an ecosystem. What is the role of consumers in an ecosystem? [04]

- Q - 5 (a)** Explain oxygen cycle in detail. [04]
(b) Compare Energy flow and Nutrient flow in an ecosystem. [05]
(c) Enlist components of Grassland Ecosystem. Classify Natural Resources. [05]

OR

- Q - 5 (a)** Define Biogeochemical cycling. Write environmental significance of nitrogen cycle. [04]
(b) Define 'Ecological Pyramid'. Explain Ecological Pyramid of Number. [05]
(c) Enlist components of Forest Ecosystem. Enlist impacts due to overexploitation of forest resources. [05]

- Q - 6 (a)** Enlist factors affecting human settlement. [04]
(b) Write ecological and economical role of forest resources. [05]
(c) Explain Optimum Population Theory. [05]

OR

- Q - 6 (a)** Enlist problems due to Overexploitation of Energy Resources. [04]
(b) Write advantages and limitations of Non- Conventional Energy Resources. [05]
(c) Explain Demographic Transition Theory in detail. [05]
